



Lecture

6

Elasticity of Demand



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The responsiveness of one variable to changes in another

When price rises, what happens to demand?

Demand falls BUT! How much does demand fall?

If price rises by 10% - what happens to demand?

We know demand will fall

By more than 10%?

By less than 10%?

Elasticity measures the extent to which demand will change

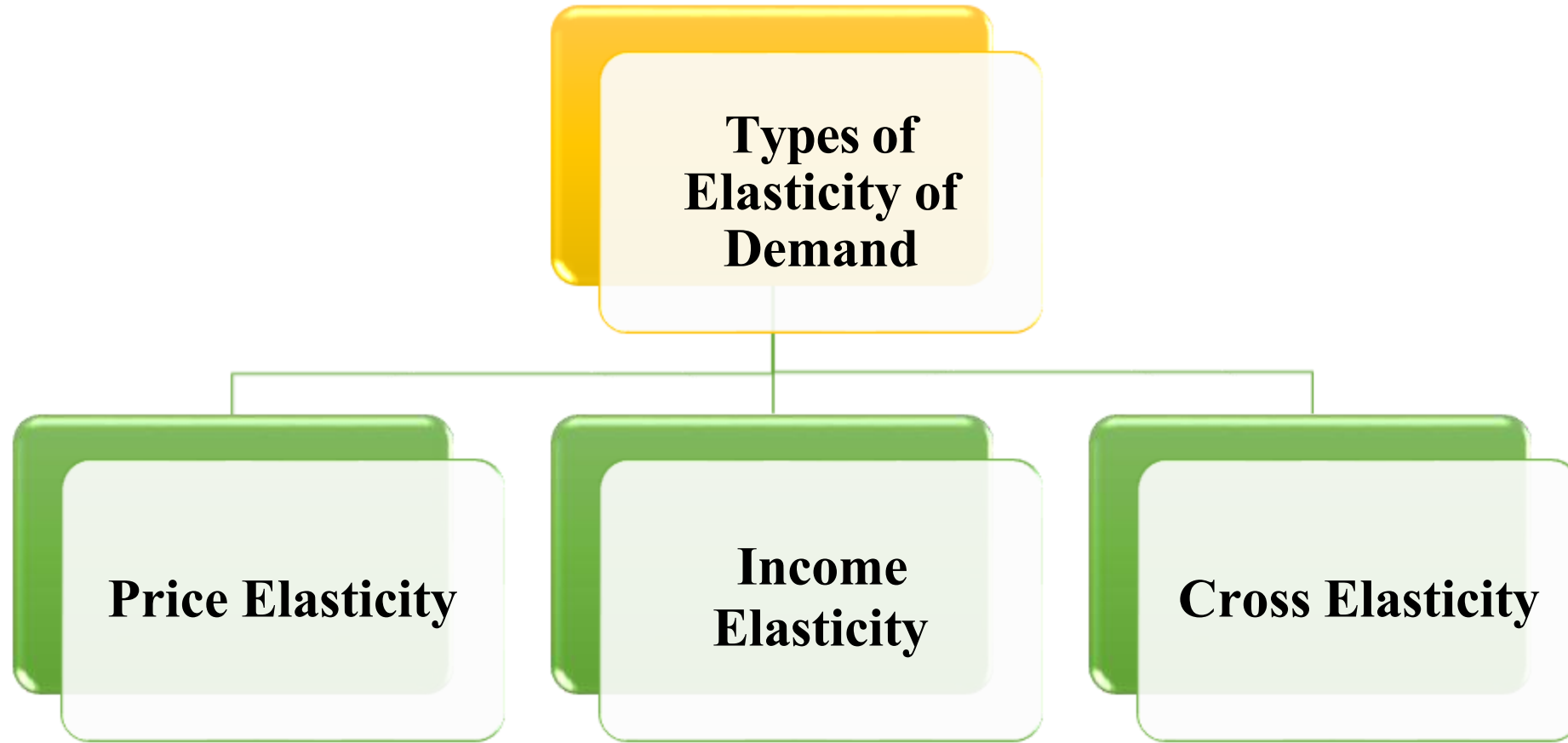


Definition of Elasticity of Demand

It is defined as the responsiveness of the quantity demanded of good to the change in price, income, and price of related good.

$$E_D = \frac{\% \text{ Change in quantity demanded}}{\% \text{ Change in any one of the determinant of demand}}$$

Types of Elasticity of Demand





Price Elasticity of Demand

- Price Elasticity of Demand expresses response of quantity demanded of a commodity to changes in its price.

$$E_p = \frac{\% \text{ Change in quantity demanded}}{\% \text{ Change in price}}$$

$$EP = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

- Where:

ΔQ = Change in quantity

ΔP = Change in price

Q = Original quantity

P = Original price

Degrees of Price Elasticity-

1. Perfectly Elastic Demand- $E=\infty$

- Consumers are **extremely sensitive** to price changes.
- At the given price, buyers will purchase **any quantity**.
- If the price increases even slightly above this level, **quantity demanded falls to zero**.
- If the price falls, quantity demanded becomes very large.

Examples

- Perfect competition in theory
- Commodities with many perfect substitutes



2. Perfectly Inelastic Demand- $E=0$

- Consumers are **completely insensitive** to price changes.
- They buy the **same quantity** at all prices.
- Price may rise or fall, but quantity demanded does not change.

Why elasticity is zero

- Quantity change is **zero**, even when price changes.

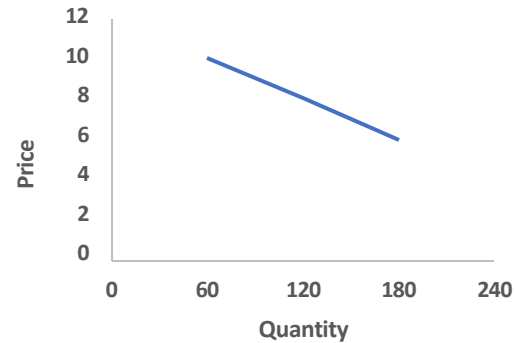
Examples

- Life-saving medicines (insulin)
- Essential goods with no substitutes in the short run



3. Unitary Elastic Demand $E=1$

- The percentage change in quantity demanded is exactly equal to the percentage change in price.
- Example:
 - Price ↓ by 10% → Quantity demanded ↑ by 10%



4. Highly Elastic or Relative Elastic Demand $E>1$

- Quantity demanded is **very responsive** to price changes.
- Percentage change in quantity demanded is **greater** than percentage change in price.

Examples

- Luxury goods
- Branded items

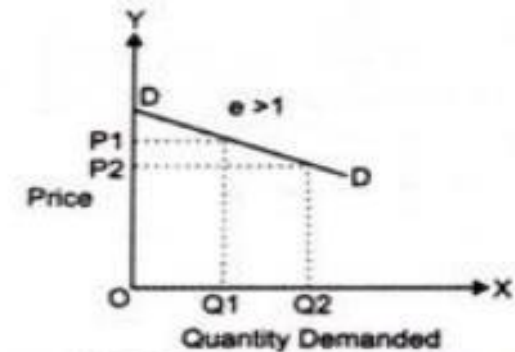


Figure-4: Relatively Elastic Demand

5. Inelastic Demand $E<1$

- Quantity demanded is **not very responsive** to price changes.
- Percentage change in quantity demanded is **less** than percentage change in price.

Examples

- Basic food items
- Fuel (in short run)

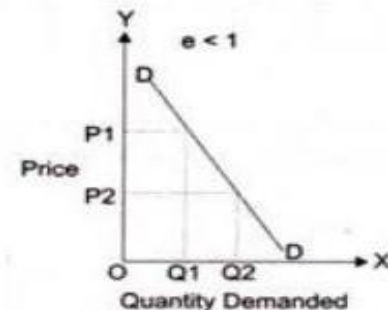


Figure-5: Relatively Inelastic Demand



Income Elasticity of Demand

- The degree of responsiveness of demand for a product to the **change in income** of a consumer is known as income elasticity.

$$E_Y = \frac{\% \text{ Change in quantity demanded}}{\% \text{ Change in income of consumer}}$$



Types of Income Elasticity of Demand

1. Income elasticity more than 1, it is luxury goods or a superior goods.

$$E > 1$$

2. Income elasticity is more than 0 but less than 1, a necessity good or a necessary good is a type of normal good

$$0 < E < 1 \quad \text{Essential goods}$$

3. Income elasticity is 0, **that an increase in income does not change the quantity demanded of the good.**

$$E = 0 \quad \text{Inexpensive and necessity goods}$$

4. Income elasticity is negative is associated with inferior goods .

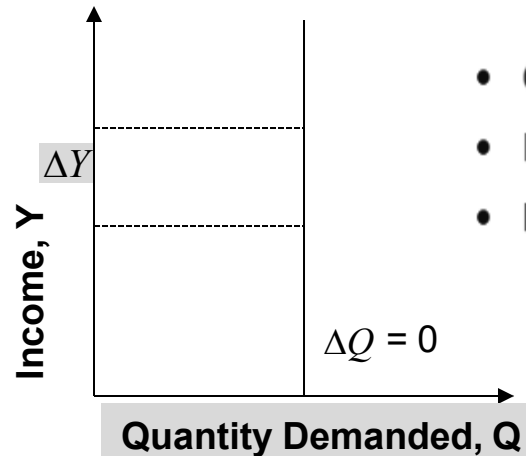
$$E < 0$$

5. Income elasticity is equal to 1

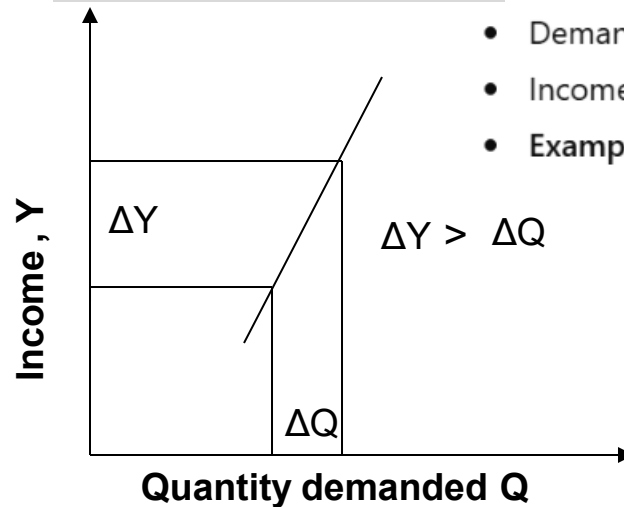
$$E = 1 \quad \text{Semi Luxury}$$

TYPES OF INCOME ELASTICITY OF DEMAND

Zero Income Elasticity of Demand ($e_Y = 0$)

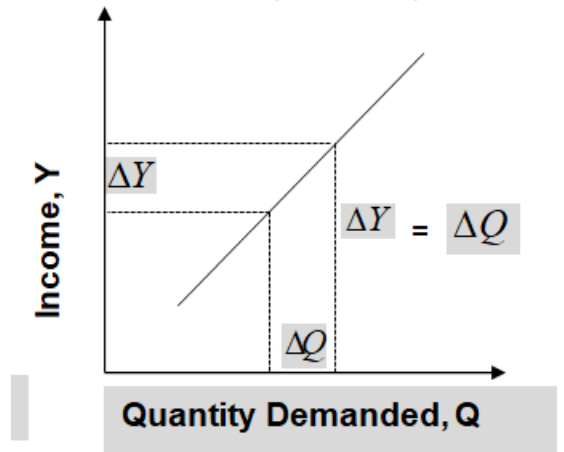


- Quantity demanded **does not change** when income changes.
- Income ↑ or ↓ → Demand remains same.
- **Example:** Salt, basic necessities.



Low Income Elasticity of Demand ($e_Y < 1$)

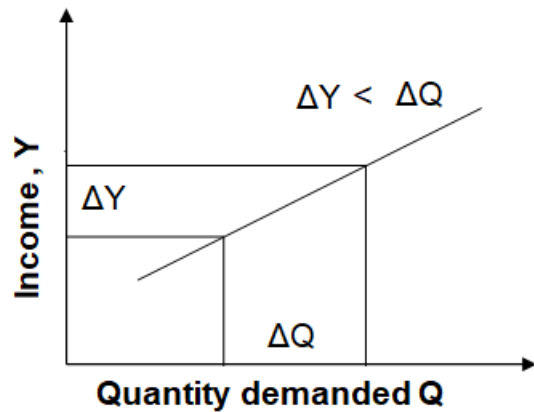
Unit Income Elasticity of Demand ($e_Y = 1$)



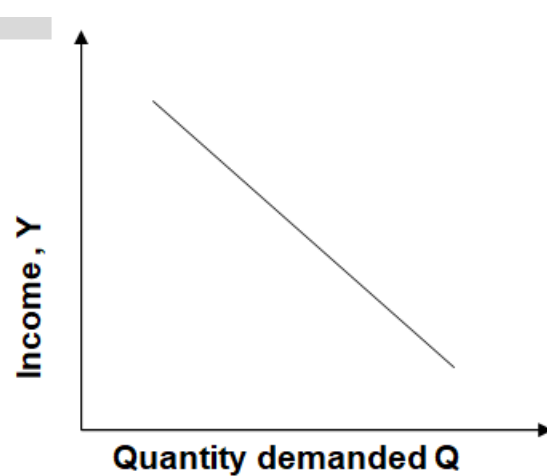
- Demand changes in the **same proportion** as income.
- Income ↑ 10% → Demand ↑ 10%.
- **Example:** Normal household goods.



High Income Elasticity of Demand ($e_Y > 1$)



- Demand increases **more than** income.
- Income ↑ slightly → Demand ↑ a lot.
- **Example:** Luxury goods (cars, ACs).



- Demand **falls** when income rises.
- Indicates **inferior goods**.
- **Example:** Cheap substitutes, low-quality goods.

Negative Income Elasticity of Demand ($e_Y < 0$)



Cross Elasticity of Demand

- A change in price of **one article** causes a change in demand of another article.

$$E_c = \frac{\textit{Proportionate change in purchase of commodity}}{\textit{Proportionate change in price of commodity}}$$

Cross Elasticity of Demand

It is the responsiveness of the demand for a **commodity X** to a change in price of **commodity Y**. It is given by

$$e_{XY} = \frac{\% \text{ change in Quantity demanded of X (} Q_X \text{)}}{\% \text{ change in price of Commodity Y.}}$$

$$e_{XY} = \frac{\Delta Q_X / Q_X}{\Delta P_Y / P_Y} = \frac{\Delta Q_X \cdot P_Y}{\Delta P_Y \cdot Q_X}$$

e_{XY} - Positive for substitutes as price and demand are in direct proportion.

e_{XY} - Negative for Complementary goods as income and demand are inversely proportional.

e_{XY} - Zero for unrelated goods.

Promotional Elasticity of Demand

The expansion of demand by means of advertisement and other promotional efforts is measured by promotional elasticity of demand.

$$e_A = \frac{\% \text{ Change in commodity X}}{\% \text{ Change in Advertisement Expenses, A.}}$$

$$e_A = \frac{\Delta Q / Q}{\Delta A / A} = \frac{\Delta Q \cdot A}{\Delta A \cdot Q}$$

1. When cross elasticity is more than 0 (Substitute goods)

$$E_c > 0$$

2. When cross elasticity is less than 0 (Complimentary goods)

$$E_c < 0$$

3. When cross elasticity is equal to 0 (Unrelated goods)

$$E_c = 0$$